EXPERIMENT NO. 15: EASYTOLL ONLINE TOLL PAYMENT SYSTEM

AIM:

To develop the EasyToll automated toll management web application allowing motorists to register vehicles, recharge eWallets, purchase advance toll passes, generate dynamic QR codes, and simulate toll booth QR scanning and pass verification.

PSEUDOCODE (IN 6 SIMPLE POINTS):

Step 1: Register motorist accounts with vehicle registration number and category (Car/Jeep, Light Commercial, Bus/Truck).

Step 2: Maintain an integrated eWallet system allowing users to recharge balance and view deduction logs.

Step 3: Provide functionality to purchase advance single-journey toll eTickets or apply for monthly recurring route passes.

Step 4: Generate dynamic encrypted QR codes encoding vehicle registration, toll plaza ID, and timestamp upon pass issuance.

Step 5: Equip toll booth operators with a QR scanner verification simulation that validates pass authenticity and deducts fare.

Step 6: Provide admin dashboard to configure plaza toll tariffs, monitor real-time vehicular traffic, and generate toll reports.

SOURCE CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>EasyToll - Electronic Toll Collection System</title>
  <style>
    body { font-family: 'Segoe UI', Tahoma, sans-serif; margin: 0; background: #f8fafc; color: #1e293b; }
    header { background: #047857; color: white; padding: 20px 35px; display: flex; justify-content: space-between; align-items: center; }
    .container { max-width: 950px; margin: 25px auto; background: white; padding: 25px; border-radius: 8px; box-shadow: 0 4px 12px rgba(4,120,87,0.1); }
    .grid-2 { display: grid; grid-template-columns: 1fr 1fr; gap: 20px; }
    .card { background: #ecfdf5; border: 1px solid #a7f3d0; padding: 20px; border-radius: 8px; }
    input, select, button { width: 100%; padding: 10px; margin: 6px 0; border: 1px solid #cbd5e1; border-radius: 6px; box-sizing: border-box; }
    button { background: #047857; color: white; font-weight: bold; border: none; cursor: pointer; }
    button:hover { background: #065f46; }
    .qr-box { border: 2px dashed #047857; padding: 20px; text-align: center; border-radius: 8px; margin-top: 15px; background: white; }
  </style>
</head>
<body>
  <header>
    <h2>EasyToll Advance Toll Pass System</h2>
    <span>eWallet Balance: Rs. <b id="walletBal">1,200</b></span>
  </header>
  <div class="container">
    <div class="grid-layout grid-2">
      <!-- User Pass Application -->
      <div class="card">
        <h3>Motorist: Buy Toll eTicket / Pass</h3>
        <label>Vehicle Registration No:</label>
        <input type="text" id="vehNo" placeholder="e.g. TN-07-BW-5521" value="TN-07-BW-5521">
```

```
            <label>Vehicle Category:</label>
            <select id="vehType" onchange="calcTollFare()">
              <option value="90">Car / Jeep / Van (Single: Rs. 90)</option>
              <option value="150">Light Commercial Vehicle (Single: Rs. 150)</option>
              <option value="310">Bus / 3-Axle Truck (Single: Rs. 310)</option>
            </select>
            <label>Pass Type:</label>
            <select id="passType" onchange="calcTollFare()">
              <option value="single">Single Trip</option>
              <option value="monthly">Monthly Route Pass (Rs. 1,800)</option>
            </select>
            <p>Payable Amount: <b>Rs. <span id="tollCost">90</span></b></p>
            <button onclick="purchaseTollPass()">Pay & Generate QR Pass</button>
          </div>

          <!-- Toll Gate Scanner Simulator -->
          <div class="card" style="background:#f0fdf4;">
            <h3>Toll Booth Operator: QR Scanner</h3>
            <input type="text" id="scanInput" placeholder="Scan or enter QR Pass Token">
            <button onclick="verifyTollAtBooth()" style="background:#0f766e;">Scan & Verify Toll Pass</button>
            <div id="boothScanResult" style="margin-top:15px;"></div>
          </div>
        </div>

        <div id="qrResultContainer"></div>
      </div>

      <script>
        let wallet = 1200;
        let issuedPasses = {};

        function calcTollFare() {
          let pType = document.getElementById('passType').value;
          let cost = pType === 'monthly' ? 1800 : parseInt(document.getElementById('vehType').value);
          document.getElementById('tollCost').innerText = cost;
        }

        function purchaseTollPass() {
          let veh = document.getElementById('vehNo').value.trim();
          let cost = parseInt(document.getElementById('tollCost').innerText);
          if(!veh) return alert('Enter vehicle registration number');
          if(wallet < cost) return alert('Insufficient eWallet balance! Please recharge.');

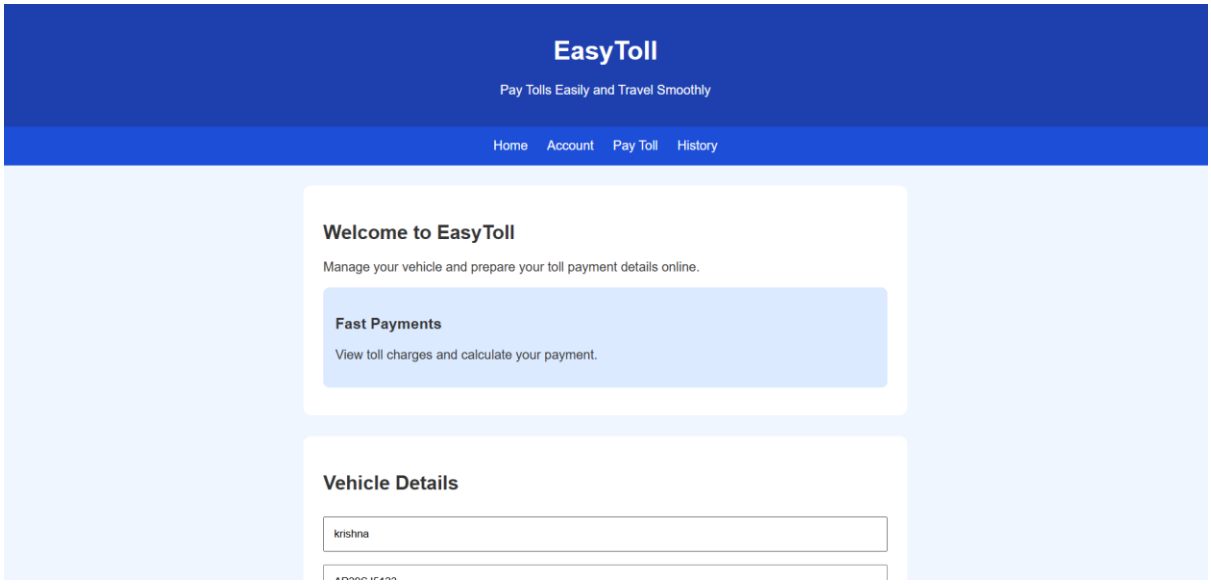
          wallet -= cost;
          document.getElementById('walletBal').innerText = wallet;
          let passId = 'TOLL-' + Math.floor(100000 + Math.random() * 900000);
          issuedPasses[passId] = { veh, cost, status: 'VALID', date: new Date().toLocaleDateString() };

          document.getElementById('qrResultContainer').innerHTML = `
            <div class="qr-box">
              <h4 style="color:#047857; margin-top:0;">Toll ePass Generated Successfully</h4>
              <p><b>Pass ID:</b> ${passId} | <b>Vehicle:</b> ${veh}</p>
              <div style="display:inline-block; border:4px solid black; padding:15px; font-family:monospace;
background:#f4f4f4;">
                [QR CODE MATRIX SIMULATION]<br>
                DATA: ${passId}||${veh}||VALID
              </div>
              <p><small>Present this QR code at toll booth optical scanner for instant barrier clearance.</small></p>
            </div>`;
          document.getElementById('scanInput').value = passId;
        }

        function verifyTollAtBooth() {
          let token = document.getElementById('scanInput').value.trim();
          let res = document.getElementById('boothScanResult');
          if(issuedPasses[token]) {
```

```
            let p = issuedPasses[token];
            res.innerHTML = `<div style="background:#dcfce7; color:#166534; padding:10px; border-radius:4px; font-
weight:bold;">
                &#10004; GREEN LIGHT: Pass ${token} Verified for Vehicle ${p.veh}. Barrier Raised!
            </div>`;
        } else {
            res.innerHTML = `<div style="background:#fee2e2; color:#991b1b; padding:10px; border-radius:4px; font-
weight:bold;">
                &#10008; RED LIGHT: Invalid or Expired Pass Token. Payment Required!
            </div>`;
        }
    }
  </script>
</body>
</html>
```

Output:



RESULT:

Thus, the EasyToll online toll payment system was successfully implemented and validated, providing eWallet deductions, advance toll ticket generation, dynamic QR pass issuance, and toll booth scanner verification.